Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc in Computer Science from Sapienza Università di Roma with experience in machine learning, NLP, computer vision, and model deployment. Built and deployed ML APIs using FastAPI, Docker, AWS ECS/Fargate, and CloudWatch, with hands-on work in latency profiling and inference optimization. Published IEEE paper and completed a research thesis on Continual Learning with Vision Transformers. Seeking Applied ML, AI Engineer, or ML Engineer roles.

EDUCATION

Sapienza Università di Roma **May 2026**

Master's in Computer Science

- **Relevant Coursework:** Deep Learning and Applied AI, NLP, Computer Vision, Distributed Systems.
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

SKILLS

- **Programming:** Python, Bash, SQL, C++
- **ML Engineering:** Machine Learning, Model Evaluation, Feature Engineering, CNNs, Transformers
- **ML Deployment:** FastAPI, REST APIs, Docker, AWS ECS, Fargate, CloudWatch
- **Frameworks:** PyTorch, TensorFlow, Keras
- **Tools:** Git, GitHub, Linux

RESEARCH EXPERIENCE

Continual Learning for Computer Vision — May 2025

- **Conducted a structured review of 60+ papers** on catastrophic forgetting, replay/regularisation methods, and TIL/DIL/CIL protocols.
- Shaped thesis research and experimental design by identifying open challenges in continual learning, guiding novel experiments to address key gaps.

Machine Learning for Air Pollution [IEEE Publication] — May 2023

- **Published IEEE ViTECoN-2023 survey paper** reviewing 6 ML models (RF, SVM, GA, LS-SVM, Decision Trees, ANN) for air quality prediction across 6 NAAQS pollutants.
- Analysed prediction-based and knowledge discovery paradigms, identified real-time monitoring and IoT integration as priority research gaps and **cited 6 times** since publication.

PROJECTS

DocDelta [Document Comparison API]

- Built a solo end-to-end semantic diff API for **legal/contract documents** using ***all-MiniLM-L6-v2*** (Sentence Transformers) + cosine similarity, detecting added, removed, and modified sentences with numerical vs. wording change classification.
- Deployed on **AWS ECS Fargate** via FastAPI + Docker. Measured **~840ms warm latency** on CPU-only inference with configurable similarity threshold and identified model preloading as key optimisation to reduce cold start from ~20s.
- Instrumented with **AWS CloudWatch** for latency and error monitoring. Stateless REST design enables horizontal scaling with zero code changes.

Lossy Text Compressor [Semantic NLP Pipeline]

- Built a semantic text compression pipeline using GTR-T5 embeddings, quantization, and vec2text-based reconstruction to study quality-compression trade-offs.
- Achieved best performance at 6-bit quantization with cosine similarity of 0.964 and ROUGE-L of 0.776, outperforming 8-, 10-, and 16-bit settings.
- Identified a non-linear relationship between compression level and reconstruction quality, with performance peaking at 6-bit and degrading beyond 8-bit.